

Waze User Churn - Analysis

Executive Summary - Ben Cochrane April 2026

Project Overview

Goal: Develop a model for predicting user churn in Waze and compare performance of different predictive models based on validation.

Result: An XGBoost Classifier model performed best and had a recall score of 16.8% in validation and 16.6% in testing, meaning it could correctly predict approximately 16% of users who would churn.

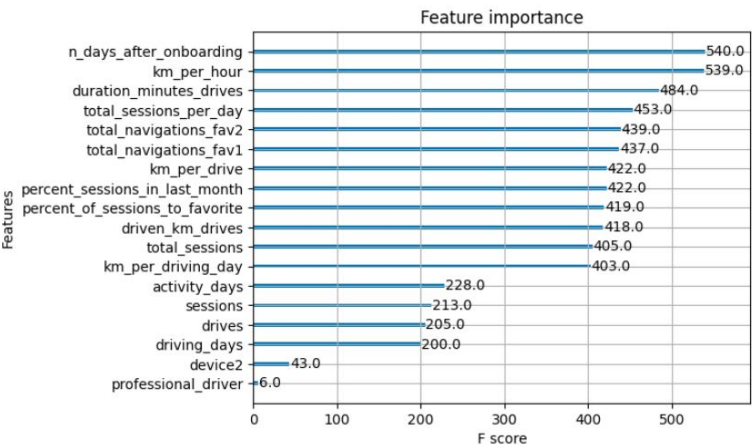
Key Insights

- Cleaned dataset with engineered features contained 14,299 observations and 18 variables
- Model performance of **binomial logistic regression, Random Forest, and XGBoost** models compared
- Recall** was selected as the performance metric to refit and optimize models on, given minimal impact of false positives
- XGBoost** performed best in validation with a recall score of **16.8%**

Details

	model	precision	recall	F1	accuracy
0	Random_Forest_Model	0.480091	0.116290	0.186928	0.820725
0	XGBoost_Classifier	0.433666	0.180692	0.254848	0.812566
0	Random_Forest_Model_Val	0.480000	0.118343	0.189873	0.820979
0	XGBoost_Classifier_Val	0.429293	0.167653	0.241135	0.812937
0	XGBoost_Classifier_Test	0.407767	0.165680	0.235624	0.809441

Model performance metrics



Next Steps

- As the recall score is fairly low, the model should not be used as a predictive model in its current state
- Given importance of engineered features shown, further feature engineering based on domain specific knowledge could improve predictive signal
- Additional data on user location, traffic conditions, drive times and other more specific aspects of app interaction could prove useful